KEY TOPIC 3 • REVISION & PRACTICE COMPANION

Medicine in 18th & 19th Century Britain

c1700–c1900: Ideas on Causes, Prevention, Hospital Care, Surgery & Public Health



Primary Visual Evidence: James Gillray, *The Cow-Pock—or—the Wonderful Effects of the New Inoculation!* (1802 satirical cartoon satirising public fears of Edward Jenner's cowpox vaccine) HISTORICAL SATIRE ARCHIVE

PUPIL NAME: _____ TARGET: _____ CLASS: _____

KEY TOPIC 3 SPECIFICATION FRAMEWORK (1HI0/11)	LESSON DONE	QUIZ [10]
Key Topic 3.1: Ideas on Causes: Pasteur's Germ Theory & Koch's Bacteriology *How did swan-neck flasks (1861) and agar cultures (1876–83) demolish spontaneous generation?*	☐	☐
Key Topic 3.2: Approaches to Prevention: Edward Jenner & The Smallpox Vaccine *How did the 1796 cowpox experiment defeat variolation, and why did opposition persist until 1852?*	☐	☐
Key Topic 3.3: Improvements in Hospital Care: Florence Nightingale & Professional Nursing *How did Scutari (1854–56), Notes on Nursing (1859), and the pavilion plan transform Victorian mortality?*	☐	☐
Key Topic 3.4: The Surgical Revolution: Simpson's Chloroform & Lister's Antiseptics *How did Simpson conquer pain, Lister conquer infection, and the 1890s introduce aseptic surgery?*	☐	☐
Key Topic 3.5: Public Health & Cholera: John Snow & The 1875 Public Health Act *How did John Snow's Broad Street pump map (1854) and the 1875 Act permanently dismantle government laissez-faire?*	☐	☐

LIVING TIMELINE • PART 1: VACCINES, ANAESTHETICS & EARLY SANITATION (1700–1854)

Instructions: Complete the timeline sketches and notes as you master each enquiry lesson.

1796 • Edward Jenner: Smallpox Vaccination & Immunology

Key Topic 3.2

Jenner inoculates James Phipps with cowpox pus from Sarah Nelmes, proving cowpox confers immunity against smallpox. Publishes findings in 1798. Despite initial Royal Society scepticism, vaccination eradicates inoculator infection risks and founds immunology.

1847 • James Young Simpson: Chloroform & Anesthesia

Key Topic 3.4

Simpson discovers the potent anaesthetic properties of chloroform in Edinburgh, ending agony during surgery. Queen Victoria's use of chloroform during childbirth (1853) popularises it, enabling complex operations despite the initial 'Black Period' of surgery.

1848 • First Public Health Act & The General Board of Health

Key Topic 3.5

Spurred by Chadwick's 1842 report and epidemic cholera, Parliament passes the permissive 1848 Act. Towns can create local boards of health to supply clean water and drainage, though lack of compulsion leaves most cities vulnerable to filth and contagion.

1854 • John Snow: Broad Street Pump & Waterborne Cholera

Key Topic 3.5

During the Soho epidemic, Snow compiles his famous spot map linking 500+ fatal cholera cases to the Broad Street pump. Removing the handle immediately halts the outbreak, mathematically disproving the miasma theory seven years before Pasteur's Germ Theory.

The 18th century relied on empirical observation; the 19th century unlocked the laboratory.

LIVING TIMELINE • PART 2: GERM THEORY, ANTISEPTICS & COMPULSORY PUBLIC HEALTH (1854–1900)

Instructions: Complete the timeline sketches and notes as you master each enquiry lesson.

1854–56 • Florence Nightingale: Hospital Sanitation at Scutari

Key Topic 3.3

During the Crimean War, Nightingale enforces rigorous cleanliness, ventilation, and fresh nutrition at Scutari hospital, slashing soldier death rates from 40% to 2%. Returns to Britain to establish the Nightingale Training School and author Notes on Nursing (1859).

1861 • Louis Pasteur: Germ Theory & Swan-Neck Flask Experiments

Key Topic 3.1

Commissioned by Lille brewers, Pasteur disproves spontaneous generation using swan-neck flasks, demonstrating that airborne microbes cause fermentation and decay. In 1861, he publishes Germ Theory, proving microscopic germs cause specific diseases in living organisms.

1865–67 • Joseph Lister: Carbolic Acid Spray & Antiseptic Surgery

Key Topic 3.4

Applying Pasteur's Germ Theory to wound sepsis, Lister uses carbolic acid sprays, dressings, and hand-washing during surgery at Glasgow Infirmary. His antiseptic techniques reduce amputee mortality from 46% to 15%, paving the way for modern aseptic operating theatres.

1875 • The Second Public Health Act: End of Laissez-Faire

Key Topic 3.5

Disraeli's government ends laissez-faire by making public health compulsory. Municipal councils are legally required to provide clean piped water, subterranean sewer drainage, street lighting, and medical officers of health, transforming urban life expectancy.

Pasteur opened the door; Lister, Koch, and Bazalgette marched through it.

Why did it take 30 years for Germ Theory to transform medical understanding in Britain?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **c1700–1850s**

Spontaneous Generation & Miasma

- Miasma: airborne decay blamed for disease
- Spontaneous generation: rotting creates microbes
- Animalcules observed without disease link
- Anti-contagionist medical orthodoxy

FOCUS CLUE
Why did doctors believe disease created germs, rather than vice versa?

2 **1861**

Louis Pasteur & Germ Theory

- Investigates beer & wine souring in Lille
- Swan-neck flask traps airborne dust
- Microbes cause decay and fermentation
- 1861: Publishes Germ Theory of decay

FOCUS CLUE
How did the swan-neck flask disprove spontaneous generation?

3 **1867–1876**

British Scepticism & John Tyndall

- Dr Charlton Bastian defends miasma theory
- Pasteur dismissed in Britain as "mere chemist"
- 1876: Tyndall lectures on dust and germs
- Joseph Lister applies theory to carbolic spray

FOCUS CLUE
Why did British doctors refuse to believe germs caused human illness?

4 **1876–1882**

Robert Koch & Specific Bacteria

- 1876: Isolates specific anthrax bacillus
- Agar jelly grows pure bacterial cultures
- Methyl violet industrial dye stains bacteria
- 1882: Discovers Mycobacterium tuberculosis

FOCUS CLUE
How did Koch's staining and culture methods prove specificity?

5 **1883–1890**

Postulates & Microphotography

- 1883: Identifies cholera microbe in Egypt/India
- High-resolution photomicrography proves facts
- 4 Postulates: scientific gold standard
- Permanent scientific defeat of miasma theory

FOCUS CLUE
Why was photomicrography crucial in winning British doctors over?

Pasteur proved that rotting soup doesn't spontaneously create germs; Koch proved which specific germs caused tuberculosis.

Question 4: Explain Why [12 marks • 15 mins]

Explain why there was rapid progress in understanding the causes of disease in the period c1860–c1890. [12 marks]

Stimulus: Louis Pasteur's Germ Theory (1861) | Robert Koch's work on bacteria *(You must also use information of your own)*

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.1). In the drawing box, sketch Pasteur's swan-neck flask experiment and Koch's agar plate. Annotate how identifying specific bacteria ended miasma theory forever! **[ ] Done**

Robert Koch photographed bacteria so clearly that British doctors could no longer blame bad smells for tuberculosis.

Continue your response to Question 4 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which ideas about the cause of disease in the late nineteenth century (c1860–c1890) were different from ideas in the Renaissance (c1500–c1700). [4 marks]

Structure your causal argument: demonstrate how Koch built directly upon Pasteur's Germ Theory to revolutionize diagnosis.

Why did Edward Jenner face fierce resistance despite discovering the smallpox vaccine?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **1721–1795**

Smallpox & Inoculation

- Smallpox killed 40,000+ Britons each year
- Lady Montagu introduced Turkish variolation
- Inoculation risked triggering deadly epidemics
- Suttonian commercial method charged high fees

FOCUS CLUE

Why was inoculation dangerous for poor urban communities?

2 **1796–1798**

Edward Jenner & The Cowpox Vaccine

- Gloucestershire dairy milkmaids immune to smallpox
- May 1796: Inoculates James Phipps with cowpox
- July 1796: Variolates Phipps—no smallpox develops
- 1798: Publishes empirical findings at own expense

FOCUS CLUE

How did Jenner use scientific experimentation to test folklore?

3 **1798–1810s**

Medical & Religious Opposition

- Royal Society refuses to publish Jenner's paper
- Dr Woodville uses contaminated hospital needles
- Clergy condemn injecting beast matter into humans
- Anti-vaccine cartoons depict cow heads sprouting

FOCUS CLUE

Why did London doctors and church leaders resist vaccination?

4 **1802–1840**

Parliamentary Funding & Adoption

- 1802 & 1807: Parliament grants Jenner £30,000
- Napoleon vaccinates entire French grand army
- 1808: Royal Jennerian Society founded
- 1840: Vaccination Act bans dangerous inoculation

FOCUS CLUE

Why did the British government abandon laissez-faire for vaccination?

5 **1852–1871**

Compulsory Acts & Anti-Vaccine League

- 1852: Vaccination made compulsory for all infants
- 1871: Vaccination officers enforce mandatory fines
- Anti-Vaccination League marches for civil liberty
- Jenner's limit: unable to explain viral mechanism

FOCUS CLUE

Why did making vaccination compulsory spark fierce public protests?

Cartoons showed vaccinated people growing cow heads and horns: proof that anti-vax memes existed long before social media.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

'Edward Jenner's development of the smallpox vaccine was the most significant breakthrough in the prevention of disease in the period c1700–c1900.' How far do you agree? [16+4 marks]

Stimulus: Inoculation | The Public Health Act (1875) *(You must also use information of your own)*

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.2). In the drawing box, sketch Jenner inoculating James Phipps with cowpox. Annotate how this empirical experiment laid the foundation for modern immunology! **[ ] Done**

Continue your response to Question 5/6 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which methods of preventing smallpox in the late eighteenth century were different from methods used in the seventeenth century. [4 marks]

How did Florence Nightingale transform hospitals from death traps into places of cure?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **c1700–1850**

Voluntary Hospitals & Pre-Modern Nursing

- Voluntary hospitals funded by private charity
- Shifted from religious rest to medical treatment
- Nurses untrained, illiterate, paid less than servants
- Dickensian "Sairey Gamp" stereotype: drunk and dirty

FOCUS CLUE
Why were early Victorian hospital wards notorious death traps?

2 **1854–1856**

Scutari & The Crimean War

- Sidney Herbert sends Nightingale and 38 nurses
- Scutari built over open sewers; filth, typhus, cholera
- Scrubbed wards, clean linen, nutritious food, drains
- Mortality rate plunged from 42% to under 2%

FOCUS CLUE
How did Nightingale's sanitary reforms drastically cut the death rate?

3 **1859**

Notes on Nursing & Sanitary Science

- 1859: Publishes bestselling *Notes on Nursing*
- Insists on light, fresh air, warmth, cleanliness, diet
- Believed in miasma; hygiene worked despite wrong theory
- Used polar area pie charts to convince MPs

FOCUS CLUE
Why did Nightingale's belief in miasma still produce effective hygiene?

4 **1860s**

The Pavilion Plan Hospital Design

- Rebuilt hospitals (e.g. St Thomas' Hospital, London)
- Separate ward blocks to isolate infectious diseases
- High ceilings, large cross-ventilating windows
- Glazed, washable wall tiles to prevent filth build-up

FOCUS CLUE
How did the pavilion plan prevent cross-infection between wards?

5 **1860–1890s**

Professional Nurse Training & Asepsis

- 1860: Nightingale Training School at St Thomas'
- Nursing established as respected middle-class career
- Matrons enforced strict hygiene and moral discipline
- Late 19th c.: Germ Theory fused with nursing hygiene

FOCUS CLUE
How did nurse training permanently raise the standard of hospital care?

Nightingale believed miasma caused disease, yet her obsession with fresh air, scrubbing, and clean drains saved thousands of lives.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why the standard of hospital care and nursing improved so significantly in the second half of the nineteenth century. [12 marks]

Stimulus: Florence Nightingale's work in the Crimea (1854–56) | The pavilion hospital plan | *(You must also use information of your own)*

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.3). In the drawing box, sketch the Pavilion Hospital plan with its separate, ventilated pavilions. Annotate how cross-ventilation prevented hospital gangrene! **[ ] Done**

Continue your response to Question 4 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which hospital care in the late nineteenth century was different from hospital care in the Medieval period. [4 marks]

Why did the conquest of pain in surgery temporarily increase the patient death rate?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **c1700–1846**

Pre-Anaesthetic Surgery: Speed & Shock

- Patients held down; trauma and shock caused heart failure
- Surgeons prized raw speed (Liston: 28-second amputation)
- 1799: Humphry Davy discovers laughing gas
- 1846: William Morton demonstrates ether in Boston

FOCUS CLUE
Why was surgical speed essential before the discovery of anaesthetics?

2 **1847–1853**

James Simpson & Chloroform

- Nov 1847: Simpson discovers chloroform at dinner
- Relieved agony of childbirth; safer than volatile ether
- 1848: Hannah Greener dies from fatal overdose
- 1853: Queen Victoria uses chloroform for Prince Leopold

FOCUS CLUE
Why did the death of Hannah Greener cause panic among surgeons?

3 **1846–1865**

The "Black Period" of Surgery

- Anaesthetics allowed surgeons to operate deeper & longer
- Surgeons wore pus-stained frock coats as badges of pride
- Unwashed hands and catgut ligatures introduced germs
- Hospital gangrene and sepsis mortality surged dramatically

FOCUS CLUE
Why did conquering surgical pain lead to higher patient death rates?

4 **1865–1870s**

Joseph Lister & Carbolic Acid Spray

- Lister reads Pasteur; links cattle sewage treatment to wounds
- 1865: Uses carbolic acid on Jamie Greenlees' compound fracture
- Glasgow Infirmary amputee mortality drops from 46% to 15%
- 1867: Publishes antiseptic technique in *The Lancet*

FOCUS CLUE
How did Lister apply Germ Theory to solve hospital gangrene?

5 **1880s–1890s**

From Antiseptic to Aseptic Surgery

- Antiseptic: kills germs present (carbolic cracked hands)
- Aseptic: excludes all microbes from operating theatre
- 1881: Robert Koch invents high-pressure steam sterilizer
- 1890s: Rubber gloves (Halsted), surgical masks, sterile gowns

FOCUS CLUE
What is the crucial difference between antiseptic and aseptic surgery?

Chloroform conquered pain, but Lister conquered the invisible enemy: the rotting microbes thriving on surgeons' unwashed hands.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why surgery became significantly safer in the period c1840–c1900. [12 marks]

Stimulus: James Simpson's discovery of chloroform (1847) | Joseph Lister's use of carbolic acid (1865) *(You must also use information of your own)*

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.4). In the drawing box, sketch Lister's carbolic donkey engine spraying mist over an open wound. Annotate the difference between antiseptic and aseptic surgery! **[ ] Done**

Continue your response to Question 4 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which surgical operations in the early nineteenth century (before 1846) were similar to surgery in the Medieval period. [4 marks]

Why did it take 20 years for John Snow's cholera discovery to change government policy?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 1831–1848

Epidemic Terror & Edwin Chadwick

- 1831: First cholera epidemic kills 50,000 Britons
- Violent vomiting, diarrhoea, blue skin; death within hours
- 1842: Edwin Chadwick links filth to poverty and disease
- 1848: Public Health Act creates local health boards (optional)

FOCUS CLUE
Why did towns refuse to set up health boards under the 1848 Act?

2 1854

John Snow & The Broad Street Pump

- August 1854: Severe cholera strikes Soho (500 die in 10 days)
- Snow plots deaths on spot map; clusters around Broad St pump
- Brewery workers drank beer, had private well—0 deaths
- Convinces parish vestry to remove the pump handle

FOCUS CLUE
How did Snow use epidemiological mapping to prove water transmission?

3 1854–1858

Board of Health & Miasma Orthodoxy

- General Board of Health and *The Lancet* reject Snow
- Orthodoxy claimed cholera arose from foul cesspool miasma
- Snow lacked bacteriological proof (Koch found cholera in 1883)
- Cesspool leaking into pump well discovered too late

FOCUS CLUE
Why did the medical establishment refuse to accept Snow's water theory?

4 1858–1865

The Great Stink & Bazalgette's Sewers

- Summer 1858: Unbearable Thames stench closes Parliament
- MPs fear miasma; fund Joseph Bazalgette's sewer network
- 1,300 miles of underground sewers divert waste to Thames estuary
- 1866: East London is only area to get cholera (unconnected)

FOCUS CLUE
Why did fear of miasma paradoxically save thousands from cholera?

5 1875

The 1875 Public Health Act: End of Laissez-Faire

- 1867 Reform Act gives working-class men the vote
- Disraeli's government passes landmark 1875 Public Health Act
- Compulsory: clean piped water, sewers, street cleaning, food checks
- Each council required to appoint a Medical Officer of Health

FOCUS CLUE
Why was the 1875 Act the definitive death blow to laissez-faire?

Snow removed the Broad Street pump handle in 1854, proving cholera was water-borne seven years before Germ Theory.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

'The work of John Snow was the main reason for improvements in public health in the nineteenth century.' How far do you agree? [16+4 marks]

Stimulus: Edwin Chadwick's 1842 Report | The Public Health Act (1875) *(You must also use information of your own)*

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 3.5). In the drawing box, sketch John Snow's Broad Street pump map and Bazalgette's sewer network. Annotate how cholera forced Parliament to abandon laissez-faire! **[ ] Done**

Continue your response to Question 5/6 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which government action on public health in the late nineteenth century was different from government action in the Renaissance. [4 marks]

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT3.1	**Q4:** Explain Why Rapid Progress in Causes c1860–c1890	p. 5–7	/ 12	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Difference (Causes: c1700 vs c1900)	p. 7	/ 4	
KT3.2	**Q5/6:** Evaluative Essay on Jenner Smallpox Vaccine c1700–c1900	p. 9–11	/ 20	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Difference (Smallpox Prevention Methods)	p. 11	/ 4	
KT3.3	**Q4:** Explain Why Hospital Care & Nursing Improved c1850–c1900	p. 13–15	/ 12	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Difference (Hospital Care: Medieval vs c1900)	p. 15	/ 4	
KT3.4	**Q4:** Explain Why Surgery Became Safer c1840–c1900	p. 17–19	/ 12	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Similarity in Surgery (Medieval vs Pre-1846)	p. 19	/ 4	
KT3.5	**Q5/6:** Evaluative Essay on John Snow & 1875 Public Health Act	p. 21–23	/ 20	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Difference in Public Health Action (1848 vs 1875)	p. 23	/ 4	
TOTAL	**Key Topic 3 Cumulative Examination Assessment Portfolio (10 Tasks)**	p. 4–23	/ 96	**Overall Grade:** _____ Teacher Target: _____

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES — Interactive Textbook • 10 Questions Per Enquiry

KT3.1	KT3.2	KT3.3	KT3.4	KT3.5
Pasteur & Koch	Edward Jenner	F. Nightingale	Simpson & Lister	Snow & 1875 Act
LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB
MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ
Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]

Sanitation, antiseptics, and vaccinations conquered disease; retrieval practice will conquer your GCSE exam.